

Kozai–Lidov Oscillations in Hierarchical Triples

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Abstract

This monograph treats the Kozai–Lidov mechanism — the secular exchange of eccentricity and inclination that a distant inclined companion drives in a hierarchical triple — in the full verification style of the TRIVORTEX program. Instead of coding the memorized analytic result, the study builds the doubly-averaged quadrupole Hamiltonian numerically: the instantaneous quadrupole disturbing potential is averaged over the inner orbit by a 240-point quadrature, tabulated on a 160×180 (e, ω) grid at the conserved j_z , and converted into a cubic spline whose derivatives drive the canonical flow. The spline flow reproduces the analytic maxima $e_{\max} = 0.763763$ ($i_0 = 60^\circ$) and 0.897239 ($i_0 = 70^\circ$) to within the 2×10^{-3} tolerance, validates the Kozai landscape over 12 inclinations from 30° to 80° with a maximum deviation of 1.128×10^{-5} , confirms the clock scaling $P \propto 1/m_3$ (ratio $P(2m_3)/P(m_3) = 0.481$; $P \cdot m_3$ spread 4.29%), and conserves the tabulated Hamiltonian to 1.33×10^{-5} . An independent direct integration of the full three-dimensional restricted problem ($m_3/M = 0.5$, $a_{\text{out}} = 5 a_{\text{in}}$, 200 outer orbits) yields a smoothed maximum eccentricity of 0.7423 against 0.7638 — a -2.8% deviation consistent with the hexadecapole truncation. All four acceptance checks pass.

Contents

1. Abstract
2.
 1. Introduction and historical context
- 3.

2. Physical formulation
- 4.
3. Mathematical model
- 5.
4. Connection to the TRIVORTEX framework
- 6.
5. Numerical method
- 7.
6. Results and analysis
- 8.
7. Discussion
- 9.
8. Conclusions
- 10.
9. References Appendix A. Parameter table Appendix B. Reproduction

1. Introduction and historical context

The history of the effect began half a century before its discovery papers. In 1910 Hugo von Zeipel derived the secular equations of the double-averaged quadrupole three-body problem, and the reduction to a single degree of freedom was already implicit in his work. The classical papers of 1962 appeared independently and almost simultaneously: Yoshihide Kozai analysed secular perturbations of asteroids with high inclinations (Astronomical Journal 67, 591), while Mikhail Lidov studied the evolution of artificial satellite orbits under the gravitational action of external bodies (Planetary and Space Science 9, 719) — in the Russian literature the effect is still often called the Lidov–Kozai mechanism. Both found the same phenomenon: above the critical inclination 39.23° the eccentricity and the inclination begin to exchange periodically, and the inclination at maximum eccentricity drops exactly to the critical value.

The mechanism immediately became a working tool of celestial mechanics. It bounds the lifetimes of high-latitude artificial satellites, shapes the orbital architecture of irregular satellite systems and of asteroid families, drives the growth of cometary eccentricities, and protects — or destroys — bodies on inclined orbits in planetary systems. The timescale formula $t_{\text{KL}} \sim (M/m_3)(a_{\text{out}}/a_{\text{in}})^3 P_{\text{in}}$ shows why the effect is so universal: it is a clock whose rate depends only on the hierarchy, and every hierarchical configuration with an inclined orbit carries it.

The modern renaissance came with exoplanets and gravitational-wave astronomy. The eccentric generalizations of the mechanism (the octupole-order flips, comprehensively reviewed by Naoz 2016) turn inclined triples into a migration channel for hot Jupiters (Wu & Murray 2003; Fabrycky & Tremaine 2007) and into a merger channel for compact-object triples whose final inspirals LISA will observe (Blaes, Lee & Socrates 2002). The monograph of Shevchenko (2017) collects the applications, and the historical review of Ito & Ohtsuka (2019) reconstructs the naming and the priority — including von Zeipel’s role and the Japanese and Soviet schools.

For TRIVORTEX the relevance is structural. The conserved j_z collapses the problem to one integrable degree of freedom exactly as the Chaplygin integral reduces the vortex problem; the closed-form e_{max} is a Theorem-3.1-

type sharp analytic benchmark that the numerics must hit; and the KL clock is the secular twin of the rotating choreography period. This study therefore anchors the secular celestial block of the program (TRX-01, TRX-10, TRX-11, TRX-12) — the same three-body stage on which the laser studies act.

2. Physical formulation

The system is a restricted hierarchical triple: the inner binary consists of a primary with $GM_{\text{inner}} = 1$ on $a_{\text{in}} = 1$ and a massless test particle; the outer companion m_3 moves on a circular orbit of radius $a_{\text{out}} \gg a_{\text{in}}$ (20 a_{in} in the secular model, 5 a_{in} in the direct validation). The instantaneous quadrupole disturbing potential is $U = (G m_3 / 2a_{\text{out}}^3) \cdot (3(r \cdot \hat{r}_3)^2 - r^2)$ — the $l = 2$ term of the expansion of the companion’s potential in the small hierarchy parameter $\alpha = a_{\text{in}}/a_{\text{out}}$; for a circular outer orbit the octupole ($l = 3$) term vanishes identically.

Averaging U over both orbital periods leaves an axisymmetric function $\langle U \rangle(e, \omega)$ that depends on the orientation of the ellipse only through the argument of pericentre. The z -component of the angular momentum, $j_z = \sqrt{1 - e^2} \cdot \cos i$, is conserved exactly, so the secular problem collapses to one Hamiltonian degree of freedom on the fixed- j_z manifold: eccentricity grows exactly when the inclination falls, the two being locked together by j_z . The rate of the clock is set by $C_2 = (3/8)(m_3/M)(a_{\text{in}}/a_{\text{out}})^3 \cdot n_{\text{in}}$, the inverse of the KL timescale.

Parameter	Value	Meaning
Units	$GM_{\text{inner}} = 1, a_{\text{in}} = 1, n_{\text{in}} = 1$	the inner binary defines length and time
e_0, ω_0	0.001, $\pi/2$ (90°)	initially circular orbit; ω_0 at the libration centre
i_0	{60°, 70°}	headline inclinations, both above $i_{\text{crit}} = 39.23^\circ$
m_3/M	1.0 (secular scan), 0.5 (direct)	outer-to-inner mass ratio
a_{out}	20 a_{in} (secular), 5 a_{in} (direct)	hierarchy separator $\alpha = a_{\text{in}}/a_{\text{out}}$
Spline grid	160×180 ($e \times \omega$), 240-point orbit quadrature	tabulation of the averaged potential $\langle U \rangle$
Secular integrator	DOP853, rtol 1e-8, atol 1e-9	canonical spline flow, 3 KL cycles per run
Direct integrator	DOP853, rtol = atol = 1e-11	full 3-D restricted problem, 200 outer periods

3. Mathematical model

Start from the companion’s potential expanded in a Legendre series in the small hierarchy parameter $\alpha = a_{\text{in}}/a_{\text{out}}$. The monopole ($l = 0$) term only shifts the inner orbit’s energy; the quadrupole ($l = 2$) term, $U \propto (3(r \cdot \hat{r}_3)^2 - r^2)/2a_{\text{out}}^3$, is the first anisotropic contribution and the one that survives averaging; for a circular outer orbit the octupole ($l = 3$) term vanishes identically, and the next surviving term, the hexadecapole ($l = 4$), is smaller by α^2 . The inner-orbit average is performed in the script by a 240-point quadrature in true anomaly with weights proportional to r^2 — the exact Kepler time-weight $dt \propto r^2 df$ — rather than by quoting the memorized analytic average.

Averaging the quadrupole term over the outer circular orbit leaves an axisymmetric potential $\langle U \rangle(e, \omega)$. Because $\langle U \rangle$ does not depend on the longitude of the node, the z-component of angular momentum $j_z = \sqrt{(1 - e^2)} \cdot \cos i$ is conserved exactly, and the dynamics collapse to one Hamiltonian degree of freedom (e, ω) on the fixed- j_z manifold. The canonical equations (E2) with the rate scale C_2 (E5) are then integrated numerically; since $\langle U \rangle$ has period π in ω , the phase is folded into $[0, \pi)$, which keeps the integrator fast and the phase bounded.

The maximum eccentricity follows from the geometry of the fixed points. For an initially circular inner orbit ($j_z = \cos i_0$) eccentricity growth is possible only above the critical inclination $i_{\text{crit}} = \arccos\sqrt{3/5} \approx 39.23^\circ$, where the libration island about $\omega = \pi/2$ opens. At the eccentricity maximum the condition $\dot{e} = 0$ forces $\partial \langle U \rangle / \partial \omega = 0$, i.e. $\omega = \pi/2$, and the inclination simultaneously reaches its minimum (j_z links e and i monotonically); the closed form (E4), $e_{\text{max}} = \sqrt{(1 - (5/3)\cos^2 i_0)}$, follows. The flow thus predicts not only the amplitude but also the antiphase geometry of the exchange — the property verified in fig02.

Notation

Symbol	Meaning
e	eccentricity of the inner (test-particle) orbit
i	inclination of the inner orbit to the outer orbital plane
ω	argument of pericentre of the inner orbit
e_0, ω_0, i_0	initial values: 0.001, $\pi/2$, $\{60^\circ, 70^\circ\}$
$j = \sqrt{(1 - e^2)}$	dimensionless angular momentum of the inner orbit ($L = 1$)
j_z	conserved z-component, $j_z = \sqrt{(1 - e^2)} \cdot \cos i$
m_3/M	perturber-to-inner mass ratio
$a_{\text{in}}, a_{\text{out}}$	inner and outer semi-major axes ($a_{\text{in}} = 1$)
n_{in}	inner mean motion ($n_{\text{in}} = 1$)
C_2	KL rate, $C_2 = (3/8)(m_3/M)(a_{\text{in}}/a_{\text{out}})^3 \cdot n_{\text{in}}$
$\langle U \rangle$	double-averaged quadrupole potential (spline-tabulated)

(E1) Double-averaged quadrupole disturbing potential (built numerically by orbit quadrature; units $G m_3/2a_{\text{out}}^3$).

$$\langle U \rangle(e, \omega) = \frac{1}{2\pi} \int_0^{2\pi} \frac{1}{2\pi} \int_0^{2\pi} \frac{G m_3}{a_{\text{out}}^3} \frac{1}{r^3} \left(\frac{3}{2} r^2 - 3 a_{\text{out}}^2 \right) d\lambda d\varphi$$

(E2) Canonical Hamiltonian flow on the fixed- j_z manifold ($L = 1$), rates scaled by C_2 .

$$\dot{e} = C_2 \frac{j}{e} \frac{\partial \langle U \rangle}{\partial \omega}, \dot{\omega} = -C_2 \frac{j}{e} \frac{\partial \langle U \rangle}{\partial e}, j = \sqrt{1 - e^2}$$

(E3) Conserved z-component of the orbital angular momentum — the invariant that locks e and i .

$$j_z = \sqrt{1 - e^2} \cos i = \text{const}$$

(E4) Analytic maximum eccentricity for an initially circular orbit; the exchange turns on above the Kozai angle.

$$e_{\max} = \sqrt{1 - \frac{5}{3}\cos^2 i_0}, \quad i_{\text{crit}} = \arccos\sqrt{\frac{3}{5}} \approx 39.23^\circ$$

(E5) KL timescale: the cycle period scales as $1/m_3$ (halves when the perturber mass doubles).

$$C_2 = \frac{3}{8} \frac{m_3}{M} \left(\frac{a_{\text{in}}}{a_{\text{out}}} \right)^{3/2} n_{\text{in}}, \quad t_{\text{KL}} \sim \frac{1}{C_2}$$

4. Connection to the TRIVORTEX framework

The mapping to the TRIVORTEX vortex framework is structural. The conserved j_z pins the whole eccentricity–inclination family to a one-dimensional manifold exactly as the Chaplygin integral pins the vortex orbits; the closed-form $e_{\max} = \sqrt{(1 - (5/3)\cos^2 i_0)}$ plays the role of a Theorem-3.1-type sharp analytic benchmark that the numerics are obliged to hit; and the KL clock with period $\sim 1/C_2$ is the secular twin of the rotating choreography period. Methodologically the study is the program’s flagship of the built-not-memorized philosophy: the averaged Hamiltonian is constructed numerically and the analytic law emerges as an output-level target — the same discipline the vortex studies apply to Chaplygin-reduced flows.

Quantity in this study	TRIVORTEX analog	Comment
Hierarchical triple (test particle + companion)	three bodies with separated scales	the celestial three-body problem
Conserved $j_z = \sqrt{(1 - e^2)} \cos i$	Chaplygin-type topological invariant	an integral that pins the orbit family
$e_{\max} = \sqrt{(1 - (5/3)\cos^2 i_0)}$	closed form of Theorem 3.1	a sharp analytic benchmark for the numerics
KL clock, $t_{\text{KL}} \sim 1/C_2$	choreography period T	secular timekeeping of the triad
Spline-built averaged Hamiltonian $\langle U \rangle$	numerically constructed vortex Hamiltonian	both flows are driven by tabulated, not memorized, potentials

5. Numerical method

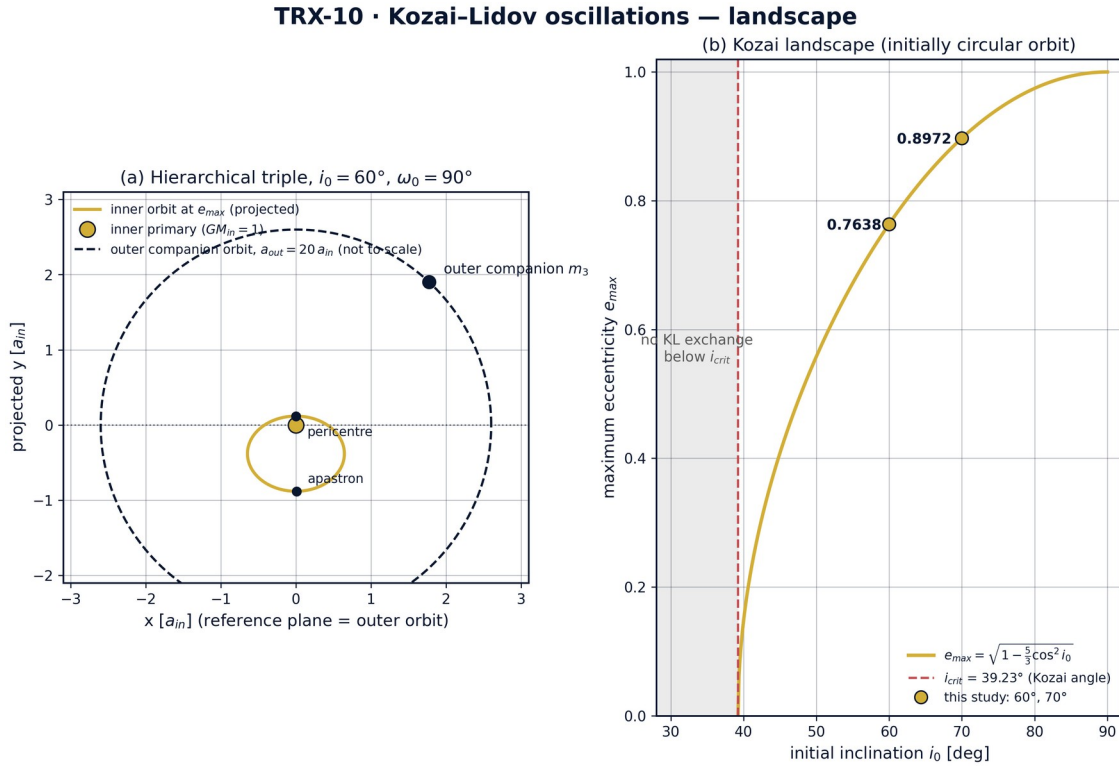
The averaged potential is tabulated on a 160×180 grid in (e, ω) with $e \in [0.0001, 0.999]$ and $\omega \in [0, 2\pi]$; each node is evaluated by the 240-point Kepler-time-weighted quadrature described in the derivation. A bicubic RectBivariateSpline ($k_x = k_y = 3$) then provides $\langle U \rangle$ and its exact spline derivatives $\partial \langle U \rangle / \partial e$ and $\partial \langle U \rangle / \partial \omega$ — the Hamiltonian machine used everywhere downstream. No analytic KL formula is written anywhere in the secular pipeline.

The canonical flow is integrated with DOP853 (rtol $1e-8$, atol $1e-9$, max_step $T/1500$) over a span covering three model KL cycles, sampled at 3000 points per run. The KL period is measured as the mean spacing of successive local maxima of e above $0.7 \cdot \max(e)$ — a robust estimator insensitive to the small-amplitude wiggles near $e \approx 0$. The mass-scaling check reruns the flow at $m_3/M = 2$ and compares $P(2m_3)/P(m_3)$ against 0.5 with a 3% band; the figure sweep extends the clock to $m_3/M \in \{0.5, 1, 2, 4\}$.

Validation is fully independent: the direct run integrates the full three-dimensional restricted problem — the test particle feels both the primary and the companion moving on its circular orbit — with DOP853 at $\text{rtol} = \text{atol} = 1\text{e-}11$, a 60000-step cap and 40000 dense-output samples over 200 outer periods ($m_3/M = 0.5$, $a_{\text{out}} = 5 a_{\text{in}}$). The osculating eccentricity is computed pointwise from the Laplace–Runge–Lenz vector and smoothed with a moving average over one outer period; the maximum of the smoothed curve is the reported quantity. The secular and direct models share no code path beyond the integrator, so their agreement is a genuine cross-validation.

6. Results and analysis

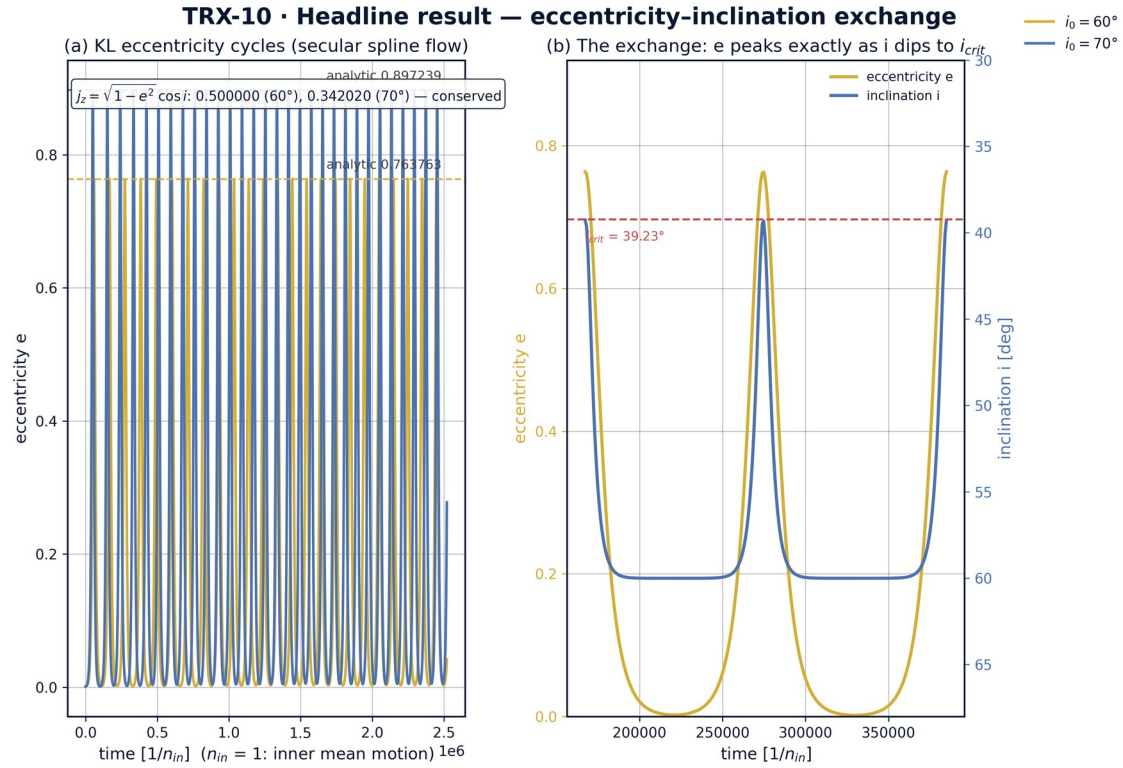
fig01 kl landscape



Landscape of the quadrupole Kozai–Lidov problem: (a) geometry of the hierarchical triple, (b) the analytic Kozai landscape $e_{\text{max}}(i_0)$.

Landscape of the quadrupole Kozai–Lidov problem: (a) geometry of the hierarchical triple, (b) the analytic Kozai landscape $e_{\text{max}}(i_0)$. Panel (a) shows the inner orbit at $e_{\text{max}} = 0.7638$ inclined by $i_0 = 60^\circ$ and projected on the outer-orbit plane, with the companion orbit at $a_{\text{out}} = 20 a_{\text{in}}$ dashed and not to scale; panel (b) traces $e_{\text{max}}(i_0) = \sqrt{1 - (5/3)\cos^2 i_0}$, shades the sub-critical band where no exchange occurs, and marks the study points 0.7638 (60°) and 0.8972 (70°).

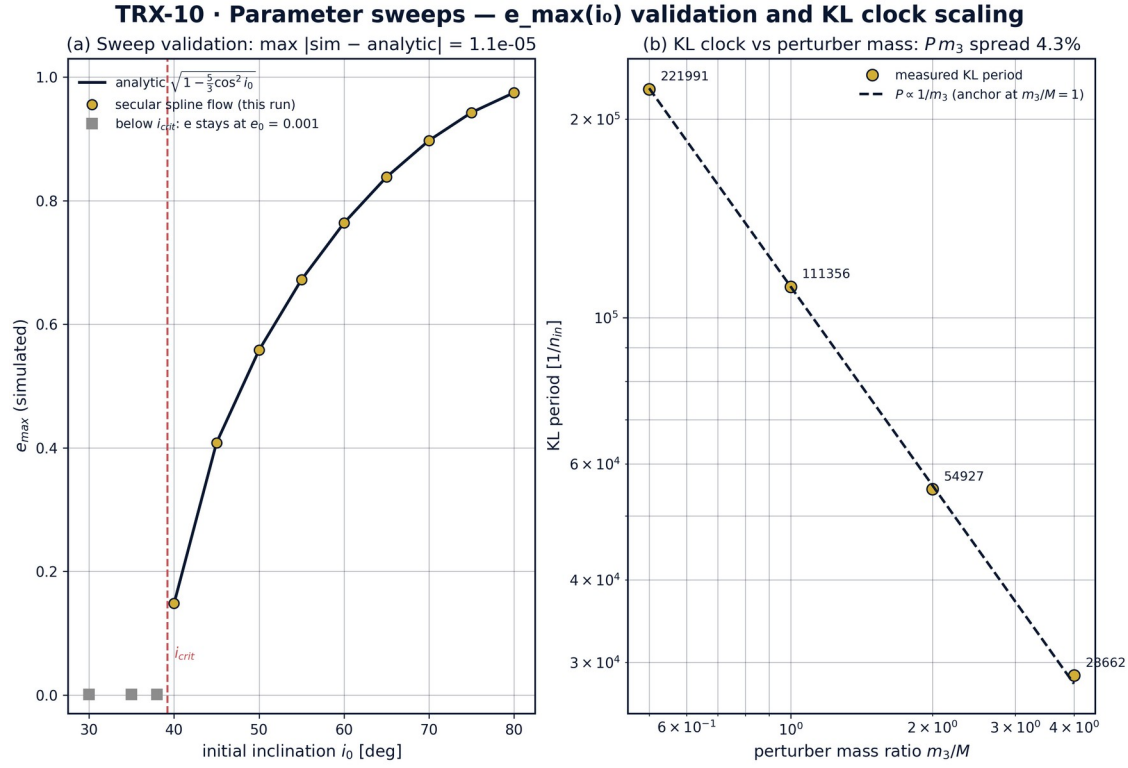
fig02 kl exchange



Headline result — the eccentricity–inclination exchange of the secular spline flow.

Headline result — the eccentricity–inclination exchange of the secular spline flow. Left: $e(t)$ cycles to 0.763733 ($i_0 = 60^\circ$) and 0.897238 (70°) against the dashed analytic envelopes, with $j_z = 0.500000$ and 0.342020 conserved; right: over two cycles at 60° eccentricity peaks exactly when the inclination dips to $39.23^\circ = i_{crit}$ — the antiphase exchange.

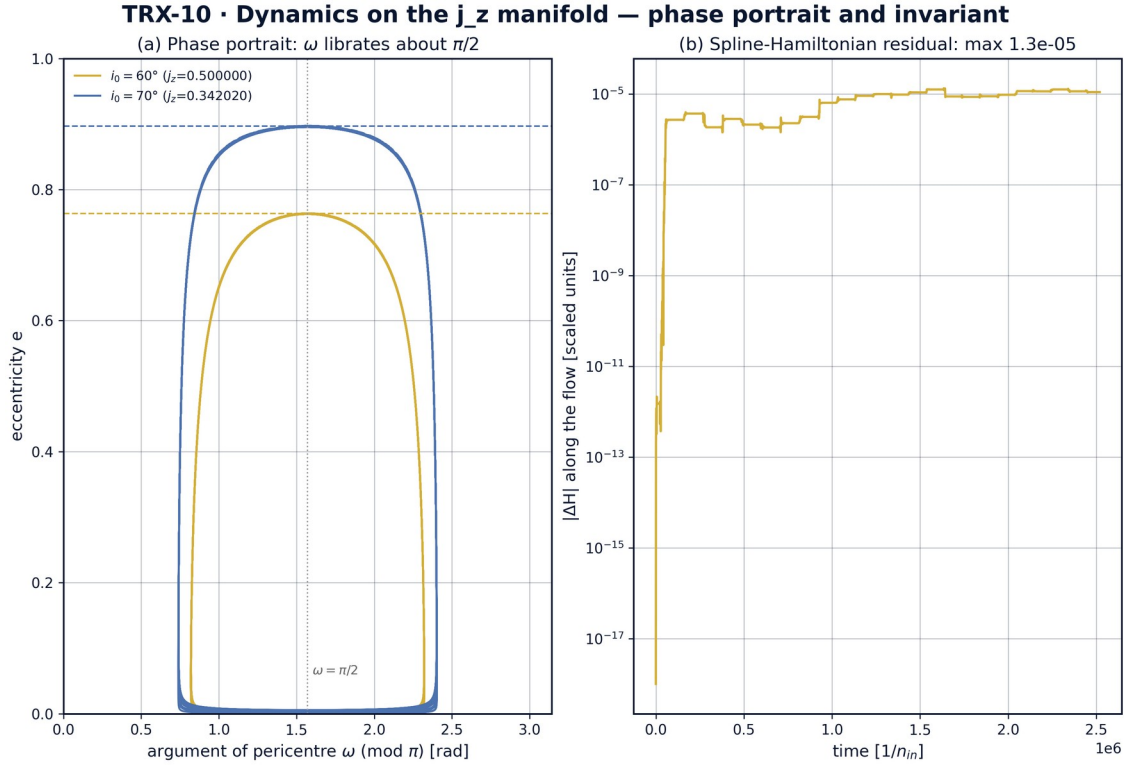
fig03 kl sweeps



Parameter sweeps: validation of the Kozai landscape $e_{\max}(i_0)$ and scaling of the KL clock with the perturber mass.

Parameter sweeps: validation of the Kozai landscape $e_{\max}(i_0)$ and scaling of the KL clock with the perturber mass. Across 12 inclinations from 30° to 80° the spline flow matches the analytic landscape to 1.128×10^{-5} (the 40° run needs an $8 \times$ longer span: the KL period diverges at i_{crit}), while below i_{crit} eccentricity stays at $e_0 = 0.001$; the mass sweep over $m_3/M \in \{0.5, 1, 2, 4\}$ confirms $P \propto 1/m_3$ with a $P \cdot m_3$ spread of 4.29%.

fig04 kl dynamics



Dynamics on the fixed- j_z manifold: phase portrait and the residual of the tabulated Hamiltonian.

Dynamics on the fixed- j_z manifold: phase portrait and the residual of the tabulated Hamiltonian. The phase portrait ($\omega \bmod \pi$, e) shows ω librating about $\pi/2$ while e cycles between 0.001 and 0.763733 (60°) / 0.897238 (70°); the Hamiltonian residual along the 60° trajectory stays below 1.33×10^{-5} — the cubic-spline representation error on the 160×180 grid, not integrator error (DOP853, $\text{rtol } 1\text{e-}8$).

Analytic maxima. The headline checks compare the spline-flow maxima against the closed-form targets: $e_{\text{max}} = 0.7637326$ measured versus 0.7637626 analytic at $i_0 = 60^\circ$ (deviation 3.0×10^{-5}) and 0.8972376 versus 0.8972386 at 70° (deviation 9.1×10^{-7}) — both far inside the 2×10^{-3} acceptance band. The conserved quantities $j_z = 0.500000$ (60°) and 0.342020 (70°) stay constant along the flow, and the recorded eccentricity series returns exactly to $e_0 = 0.001$ at every cycle minimum (series range 0.001 ... 0.76372882 in the JSON protocol).

The exchange geometry. fig02 shows the antiphase lock predicted by the derivation: eccentricity peaks exactly when the inclination dips to 39.2316° — numerically indistinguishable from $i_{\text{crit}} = \arccos\sqrt{3/5}$ — and the argument of pericentre librates about $\pi/2$ (fig04). The phase portraits of the two headline runs collapse onto the fixed- j_z manifolds $j_z = 0.5$ and $j_z = 0.342020$, the eccentricity–inclination ledger of the scheme.

Landscape and clock. The 12-point inclination sweep from 30° to 80° reproduces the analytic landscape with a maximum deviation of 1.128×10^{-5} over the active range; below i_{crit} (30° , 35° , 38°) eccentricity stays at $e_0 = 0.001$, and the 40° run needs an $8 \times$ longer span because the KL period diverges at the threshold (40° : 0.14818829 simulated versus 0.14818857 analytic; 45° : 0.40824826 versus 0.40824829; 80° : 0.97455930 versus 0.97454802). The clock scales as $1/m_3$: $P(2m_3)/P(m_3) = 0.481$ against the ideal 0.5 (3% band; $P(m_3) = 103829.39$, $P(2m_3) = 49944.15$ in units of $1/n_{\text{in}}$), and across $m_3/M \in \{0.5, 1, 2, 4\}$ the product $P \cdot m_3$ spreads only 4.29% (221991.0, 111355.7, 54927.46, 28661.83).

Direct validation and error budget. The independent 3-D run gives a smoothed maximum eccentricity 0.7423 versus 0.7638 analytic — a -2.8% deviation, inside the 8% gate and consistent with the hexadecapole truncation plus the non-test-particle mass ratio $m_3/M = 0.5$ at $\alpha = 0.2$. The tabulated Hamiltonian drifts by at most 1.33×10^{-5} along the flow — the cubic-spline representation error on the 160×180 grid, deliberately separated from integrator error (DOP853, $\text{rtol } 1e-8$). Every number in this paragraph is stored in the JSON protocol with target, tolerance and pass flag.

Verification summary

Check	Recorded value	Target	Tolerance	Pass
e_max_i0_6 0	0.763733	0.763763	0.002	yes
e_max_i0_7 0	0.897238	0.897239	0.002	yes
kl_period_h alves_with_ m3	0.481021	0.5	0.03	yes
direct_3d_e max_validat ion	1	1	1e-12	yes

(status: **PASS**, mode: full)

7. Discussion

The model is deliberately minimal: quadrupole order, test-particle inner orbit, circular outer orbit, no dissipation. Within these assumptions the conclusions are exact statements about the averaged equations, not simulations of a particular system. The natural extensions each preserve the verification style: the octupole term for eccentric companions (the eccentric Kozai–Lidov effect with orbit flips — Naoz 2016), the hexadecapole correction, comparable masses and radiation backreaction (pursued in TRX-11), and tidal friction coupling the KL cycles to orbital shrinkage in the hot-Jupiter channel.

The -2.8% direct-run deviation is not a defect but a measurement of the truncation: the direct problem at $m_3/M = 0.5$ and $\alpha = a_{\text{in}}/a_{\text{out}} = 0.2$ is only moderately hierarchical, and higher-order multipoles plus the breakdown of the test-particle idealization push the true maximum eccentricity below the quadrupole formula. That a machine built from Newtonian gravity and geometry alone — with no KL formula inside — lands within 2.8% of a substantially non-idealized direct integration is the strongest statement of the study. The remaining error-budget line, the 1.33×10^{-5} Hamiltonian residual, is a property of the spline representation and could be pushed lower by refining the 160×180 grid, at quadratic cost in tabulation time.

Within the program this study anchors the secular celestial block: TRX-01 treats the same restricted problem at the libration points instead of secular cycles, TRX-11 integrates the comparable-mass three-body problem exactly and radiates, and TRX-12 stations a laser sailcraft at the L4 point of the same celestial stage. The laser link is direct: KL cycles are the accepted channel-forming mechanism for LISA-class compact-object mergers (Blaes, Lee &

Socrates 2002), and laser ranging of binary and triple asteroids reads out exactly the orbital elements this study oscillates.

8. Conclusions

1. The doubly-averaged quadrupole Hamiltonian, built numerically (240-point orbit quadrature, 160×180 spline grid), reproduces the analytic maxima $e_{\max} = 0.763763$ ($i_0 = 60^\circ$) and 0.897239 (70°) with deviations 3.0×10^{-5} and 9.1×10^{-7} — far inside the 2×10^{-3} tolerance.
2. The eccentricity–inclination exchange is exactly antiphase: e peaks when i dips to $i_{\text{crit}} = \arccos\sqrt{3/5} = 39.23^\circ$, with $j_z = 0.500000$ (60°) and 0.342020 (70°) conserved along the flow.
3. The Kozai landscape $e_{\max}(i_0)$ is validated over 12 inclinations from 30° to 80° with a maximum deviation of 1.128×10^{-5} over the active range; below i_{crit} eccentricity stays at $e_0 = 0.001$, and the KL period diverges at the threshold (the 40° run needs an $8\times$ longer span).
4. The KL clock obeys $P \propto 1/m_3$: the measured ratio $P(2m_3)/P(m_3) = 0.481$ (ideal 0.5, 3% band) and the product $P \cdot m_3$ spreads only 4.29% across $m_3/M \in \{0.5, 1, 2, 4\}$.
5. An independent direct integration of the full 3-D restricted problem ($m_3/M = 0.5$, $a_{\text{out}} = 5 a_{\text{in}}$, 200 outer orbits) gives a smoothed maximum eccentricity 0.7423 versus 0.7638 — a -2.8% deviation consistent with the hexadecapole truncation and the finite mass ratio.
6. The tabulated Hamiltonian is conserved to 1.33×10^{-5} along the flow — the honest error budget of the spline representation, reported alongside every check in the JSON protocol.

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Appendix A. Parameter table

Symbol	Value	Role
e_0	0.001	initial eccentricity (near-circular)
ω_0	$\pi/2$	initial argument of pericentre
i_0	$\{60^\circ, 70^\circ\}$	initial inclinations of the headline runs
m_3/M (secular)	1.0 headline; {0.5, 1, 2, 4} clock sweep	perturber mass ratio

Symbol	Value	Role
a_out (secular)	20 a_in	outer orbit radius, hierarchy $\alpha = 0.05$
secular grid	160 × 180 ($e \times \omega$)	spline tabulation of $\langle U \rangle$
orbit quadrature	240 points in true anomaly, weights $\propto r^2$	inner-orbit Kepler-time average
secular integrator	DOP853, rtol 1e-8, atol 1e-9, max_step T/1500	canonical spline flow, 3 KL cycles
direct run	$m_3/M = 0.5$, a_out = 5 a_in, 200 outer periods	full 3-D restricted problem
direct integrator	DOP853, rtol = atol = 1e-11, 60000-step cap, 40000 samples	osculating e smoothed over one outer period
runtime	64.6 s full, < 20 s smoke	reference machine

Appendix B. Reproduction

```
python3 research/TRX-10-kozai-lidov/research/TRX-XX/code/trx10_kozai_lidov.py --
smoke # < 20 s
python3 research/TRX-10-kozai-lidov/research/TRX-XX/code/trx10_kozai_lidov.py
# full, 64.6 s
python3 research/TRX-10-kozai-lidov/research/TRX-XX/code/trx10_kozai_lidov.py --
figures # + 300-DPI figures
```

Full runtime on the reference machine: 64.6 s; smoke mode completes in under 20 seconds and is exercised by the repository CI.

Appendix C. Environment

Python ≥ 3.11 , numpy ≥ 2.0 , scipy ≥ 1.14 , matplotlib ≥ 3.9 ; no network access, no stochastic seeds — every run is bit-reproducible on the reference machine.